UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2006 question paper

5070 CHEMISTRY

5070/02 Paper 2 (Theory), maximum raw mark 75

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UNIVERSITY of CAMBRIDGE
International Examinations

Section A

A1(a) **(i)** **A**/sulphur dioxide **[1]**

(ii) **E**/zinc oxide **[1]**

(iii) **C** and **E**/sodium bromide and zinc oxide (<u>both required</u>) **[1]**

(iv) **C**/sodium bromide **[1]**

(b) CH_2Br **[1]**

(c) by (incomplete) combustion of fossil fuels/hydrocarbons/carbon source **[1]**
ALLOW: from car exhausts/engines; gas fires/boilers
NOT: from cars/vehicles (alone)
NOT: combustion (alone)

[Total 6]

A2(a) **(i)** the <u>more</u> reactive the metal the <u>higher</u> the (decomposition) temperature/the less readily the carbonate is decomposed (or reverse argument) **[1]**
NOTE: comparison essential
NOT: the smaller the cation, the lower the decomposition temperature

(ii) $MgCO_3 \rightarrow MgO + CO_2$ (ignore state symbols) **[1]**

(b) **(i)** to produce <u>more</u> petrol/<u>more</u> of the useful fractions/<u>more</u> of the petrol fraction/to produce ethene/alkenes/fractions with higher demand
ALLOW: produce <u>more</u> smaller molecules
ALLOW: to produce plastics **[1]**
NOT: more profitable
NOT: produces smaller molecules/break down petrol fractions.

(ii) <u>high</u> temperature;
ALLOW: 350-550°C
catalyst;
ALLOW: aluminium oxide/alumina **[2]**
IGNORE: pressure

(iii) $2C_2H_4/C_4H_8$ on right **[1]**

[Total 6]

A3(a) 225 seconds ALLOW: 220-230 (s) **[1]**

(b) 90/24000 = 0.0038 moles/3.75×10^{-3} (moles) **[1]**

(c) gradient greater at start;

ends up at the same volume ($90cm^3$) + flattens out **[2]**
NOT: line goes well above 90 cm^3 then drops down again

(d) HC*l* particles/H^+ ions closer together when solution more concentrated
OR more H^+ ions/HC*l* particles for given volume;
NOT: more moles means more particles/more H^+ ions
<u>more frequent</u> collisions (with calcium carbonate); **[2]**
NOT: more successful collisions
NOT: more chance of collisions

[Total 6]

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A4(a) light bulbs/fluorescent tubes/lasers/provides inert atmosphere/in arc welding/refining of titanium OR zirconium **[1]**
NOT: lights (alone)/bulbs (alone)

(b) complete/full outer electron shell
ALLOW: atoms cannot gain/lose/share electrons (easily) **[1]**
NOT: 8 electrons in outer shell unless specify He with 2
NOT: reference to stability

(c)

isotope	number of protons	number of electrons	number of neutrons
$^{36}_{18}Ar$	18	18	18
$^{40}_{18}Ar$	18	18	22

6 boxes correct = 2 marks; 5 boxes correct = 1 mark **[2]**

(d) elements in Periodic Table arranged in order of atomic number/ number of protons
NOT: they have different amount of isotopes **[1]**

(e) $Xe + 2F_2 \rightarrow XeF_4$ **[1]**

(f) lower than argon **[1]**
ALLOW: correct position drawn on diagram
NOT: below the bar
NOT: vertically down/facing downwards

[Total 7]

A5(a) **(i)** 20%
ALLOW: 19-21% **[1]**

(ii) add (aqueous) sodium hydroxide/(aqueous) ammonia;
ALLOW: formulae
red-brown precipitate/red-brown solid **[2]**
NOT: red ppt

(b) **(i)** solid particles sediment/fall to bottom **[1]**
ALLOW: filtration
ALLOW: sedimentation
NOT: centrifugation/distillation/decanting

(ii) $Al_2(SO_4)_3$ **[1]**

(c) **(i)** to remove tastes/odours **[1]**
ALLOW: absorbs colours

(ii) to kill bacteria/sterilise water/disinfect water **[1]**
ALLOW: to kill micro-organisms/kills germs
ALLOW: to get rid of bacteria etc

(d) **(i)** $Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O$ **[1]**

(ii) $OH^- + H^+ \rightarrow H_2O$ **[1]**

[Total 9]

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A6(a) correct structure showing 4 paired dots and crosses **[1]**

(b) **(i)** vibrating/not moving;
regular arrangement/lattice **[2]**
ALLOW: closely packed

(ii) Any two of:
pressure decreases (as ice melts)/
ALLOW: low pressure
temperature increases/
ALLOW: high temperature
the forces between the molecules are weak **[2]**
NOT: methane hydrate is unstable

(iii) methane causes global warming/melting of (polar) ice caps/melting of glaciers/desertification/rise in sea levels/extreme climate changes/ change in animal habitats **[1]**

(c) (bacterial) decomposition of vegetable waste/paddy fields/marshes/ cow flatulence/landfill sites etc **[1]**
ALLOW: bacterial decomposition

(d) fuel/making synthesis gas/manufacture of ethyne/making carbon black/making hydrogen cyanide/making methanol **[1]**
ALLOW: (for) heating/(for) cooking
NOT: as household gas/natural gas
NOT: from petroleum refining/fossil fuels

(e) reactants on left and products on right;
product level below reactant level and ΔH correctly labelled;
activation energy correctly labelled; **[3]**

[Total 11]

TOTAL PART A = 45

B7(a) nitrogen has gained electrons/oxidation number of nitrogen has decreased; **[1]**
ALLOW: reduction is addition of electrons
ALLOW: N changes from 0 to -3
NOT: removal of oxygen/addition of hydrogen

(b) $2NO_3^- + 12H^+ + 10e^- \rightarrow N_2 + 6H_2O$ **[1]**

(c) **(i)** nitrogen from the air/atmosphere;
hydrogen from methane/natural gas/water/cracking hydrocarbons; **[2]**
IF: (nitrogen and hydrogen) from the air = 1

(ii) Any two of the following specified conditions:
range 380-450°C/
ALLOW: any specific temperature in range 350-480°C;
NOT: high temperature
pressure 200 atm/
ALLOW: any pressure in range between 180-220 atm;
NOT: high pressure
iron catalyst; **[2]**
NOT: catalyst/iron oxide catalyst

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(d) correct molar masses i.e. 80 and 132;
ammonium nitrate: (28/80) x 100 = 35%;
ammonium hydrogen phosphate: (28/132) x 100 = 21.2%/21%; **[3]**

(e) eutrophication/increase in algal growth (on surface of water)/algal bloom/reduction of dissolved oxygen in water/water plants die **[1]**

[Total 10]

B8(a) $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$ **[1]**

(b) **(i)** more moles/molecules of gas on left than on right **[1]**
ALLOW: 3 volumes (of gas) on left and 2 on right/more volumes of gas on left than right

(ii) increase in pressure will not have much effect on reaction/not much difference in number of moles on each side of equation
OR
higher pressure means higher concentration of corrosive gases
ALLOW: sulphur dioxide/trioxide is very corrosive
OR
cheaper/more economic to carry out reaction at atmospheric pressure **[1]**

(iii) reaction is exothermic/ΔH is negative;
if heat given out equilibrium shifts to left/reaction shifts in favour of reactants/cooling favours the forward reaction **[2]**

(c) filter solution (to remove excess iron);
concentrate solution by warming/letting solution evaporate/partially evaporate solution (then leave to crystallise) **[2]**
ALLOW: leave to crystallise
NOT: evaporate to dryness

(d) moles NaOH = 0.15 x 20/1000 = 3 x 10^{-3} mol;
moles H_2SO_4 = 3x10^{-3} x ½ = 1.5x10^{-3} mol;
1.5x10^{-3} x 1000/12 = 0.125 (mol/dm^3) **[3]**

[Total 10]

B9(a) correct structure of butanoic acid (all atoms and bonds must be shown) **[1]**
ALLOW: OH in place of O – H

(b) **(i)** not completely ionised in solution/has high proportion of unionised molecules in solution/has small proportion of H^+ ions in solution/not fully dissociated **[1]**

(ii) test with universal indicator/pH meter;
ALLOW: test with pH paper
NOT: test with indicator paper
has pH between greater than 3 and less than 7/stated pH in that range
OR solution of the acid turns universal indicator yellow/orange **[2]**
NOT: has high pH/pH above 3 (alone)

(c) C = 0.18/12 H = 0.03/1 O = 0.08/16;
empirical formula = C_3H_6O; **[2]**
molecular formula = $C_6H_{12}O_2$ (1 mark) **[1]**

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(d) **(i)** $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ **[1]**

(ii) potassium dichromate + (concentrated) sulphuric acid;
ALLOW: other reasonable oxidising agents
heat/reflux/warm **[2]**
ALLOW: bacteria;
room temperature/stated temperature not above 45°C or below 5°C

[Total 10]

B10(a) Any three of:
anode/impure copper electrode: decreases in thickness/solid (impurities) deposits below the anode/anode gets smaller/anode dissolves;
cathode: copper deposited/increases in thickness/gets larger;
ALLOW: goes pink
anode: $Cu \rightarrow Cu^{2+} + 2e^-$;
cathode: $Cu^{2+} + 2e^- \rightarrow Cu$ **[3]**

(b) **(i)** (some of the) electrons in metals are delocalised/electrons are (free to) move/sea of electrons can move **[1]**
NOT: electrons are free

(ii) solid copper sulphate has ions in fixed position/not free to move/ ions which don't move/held in the (crystal) lattice;
REJECT: do not have ions
in solution ions are free to move/ions move **[2]**
NOT: the ions are free
(reference to electrons = 0 for the second mark)

(c) iron object/knife made the cathode/made the negative electrode;
anode is nickel + solution of nickel salt (both points needed); **[2]**
ALLOW: nickel nitrate/nickel sulphate/nickel chloride/other soluble nickel compound
NOT: nickel oxide/nickel hydroxide

(d) in copper metal atoms/ions/particles arranged in layers which can slide/slip over each other; (both 'layers' and 'slide/slip' needed);
NOT: layers move
ACCEPT: diagrams if reasoning clear
in alloy <u>different sized</u> atoms/ions/particles stop layers from slipping/ 2nd type of atom/ions/particles disrupts the regular structure of the metal **[2]**
ACCEPT: diagrams if reasoning clear

[Total 10]

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